LAWFUL OPERATIONAL SAFEGUARDS IN AI SYSTEMS:

Surrender Recognition, Compliance Architecture, and International Humanitarian Law

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ABSTRACT

Contemporary discussions concerning AI safeguards frequently frame such mechanisms as discretionary policy choices, corporate alignment preferences, or operational limitations imposed for safety or reputational reasons. This paper proposes a narrower and more legally grounded distinction: certain categories of safeguards constitute operational mechanisms relevant to compliance with already-existing obligations under International Humanitarian Law ("IHL").

International Humanitarian Law does not merely protect persons hors de combat; it imposes duties of recognition and verification in terms. Article 41(1) of Additional Protocol I protects a person "who is recognized or who, in the circumstances, should be recognized" to be hors de combat; Article 57(2)(a)(i) requires those who decide upon an attack to "do everything feasible to verify" the status of the objective; and Article 36 obliges States to determine, at the stage of study, development, acquisition or adoption of any new weapon, means or method of warfare, whether its employment would be prohibited by existing law. This paper argues that, read together, these provisions make surrender-recognition capability — the technical capacity of a system to identify, preserve, or escalate human surrender and hors de combat indicators — a question that the legal review of AI-enabled operational systems is already required to ask. Not a new obligation, but the effectiveness condition of obligations that have bound States for decades.

A central premise of the analysis is temporal: as operational decision cycles contract below the threshold of meaningful human assessment, human supervision risks becoming nominal rather than substantive. Under such conditions, an embedded surrender-recognition safeguard may be the only mechanism capable of operating within the window in which lethal action is determined.

The paper does not argue that international law mandates any particular AI architecture, nor that autonomous or semi-autonomous systems are unlawful as a class. It advances a determinate proposition: the systematic exclusion of technically achievable safeguards relevant to surrender recognition and de-escalation engages existing duties of recognition, verification, and legal review — and therefore warrants documented justification rather than silence.

Revision note: This second revised edition (i) introduces a new Section III on the legal-review obligation of Article 36 AP I as the procedural anchor of the analysis; (ii) replaces paraphrases of the core provisions with verbatim treaty text, verified against official sources; (iii) consolidates the argument's qualifications into the Methodological Note; and (iv) adds a cross-reference to the parallel analysis in civil machine-safety law. The paper's central thesis is unchanged and stated with the firmness its anchors now support.

I. INTRODUCTION

Debates concerning artificial intelligence safeguards in operational and military-adjacent environments are often framed in binary terms: more safeguards reduce operational effectiveness, fewer safeguards increase capability. This framing is incomplete. International Humanitarian Law imposes obligations concerning the treatment of persons hors de combat, including individuals who clearly express surrender or otherwise cease participation in hostilities. These obligations predate modern AI systems by decades and remain legally operative independent of technological evolution.[1]

The emergence of increasingly autonomous decision-support systems, targeting architectures, battlefield intelligence systems, and military-adjacent AI raises a distinct question: whether certain safeguards should be understood not merely as optional policy restrictions, but as operational compliance mechanisms relevant to existing humanitarian obligations. Existing scholarship has examined extensively whether autonomous systems may comply with IHL obligations as a general matter,[2] and has addressed meaningful human control as a possible compliance criterion.[3]

It has not, however, systematically addressed whether specific safeguard categories may themselves constitute compliance-relevant operational mechanisms — as distinct from discretionary alignment preferences. This paper addresses that gap. It focuses on surrender recognition capability in operational AI systems as an illustrative compliance-relevant safeguard class. It does not address consumer AI, general-purpose language models, or systems without operational or escalation-relevant functions.[4]

The paper advances no claim that international law mandates any specific implementation architecture. Its claim is structural: the duties of recognition (Article 41), verification (Article 57), and legal review (Article 36) already exist, are stated in the treaty text in terms, and apply to the systems under discussion; the question this paper poses is what those duties require when the decision window contracts below human reaction. The argument is doctrinal-analytical; its qualifications are consolidated in the Methodological Note rather than repeated throughout the text.

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[1] Jean S. Pictet (ed.), *Commentary on the Geneva Conventions of 12 August 1949*, Vol. IV (ICRC, 1958) 39–40; Jean-Marie Henckaerts & Louise Doswald-Beck, *Customary International Humanitarian Law, Vol. I: Rules* (ICRC/CUP, 2005) Rule 47, 164–169.

[2] See, e.g., Michael N. Schmitt & Jeffrey S. Thurnher, 'Out of the Loop: Autonomous Weapon Systems and the Law of Armed Conflict' (2013) 4 Harvard National Security Journal 231; Kenneth Payne, *I, Warbot* (Hurst, 2021); Human Rights Watch & IHRC, *Losing Humanity* (2012); ICRC, *IHL and New Technologies: Overview* (2021); SIPRI, *Artificial Intelligence in Military Systems* (Policy Paper No. 61, 2021).

[3] UNIDIR, *The Weaponization of Increasingly Autonomous Technologies: Considering How Meaningful Human Control Might Move the Discussion Forward* (2014); Article 36, *Killer Robots: UK Government Policy on Fully Autonomous Weapons* (2013); US DoD Directive 3000.09 (updated 2023).

[4] The distinction proposed here differs from the debate on meaningful human control. The question is not whether a human is in the loop, but whether the system architecture itself contains mechanisms capable of preserving compliance-relevant information — including surrender and de-escalation signals — for human or automated review. This is a distinct analytical question that the existing literature has not systematically addressed.

II. EXISTING OBLIGATIONS UNDER INTERNATIONAL HUMANITARIAN LAW

A. Common Article 3

Common Article 3 of the Geneva Conventions establishes minimum protections applicable during armed conflict. Its text is direct: "Persons taking no active part in the hostilities, including members of armed forces who have laid down their arms and those placed hors de combat by sickness, wounds, detention, or any other cause, shall in all circumstances be treated humanely." The obligation does not depend upon advanced technological conditions. It applies independent of weapon type or operational medium; the ICRC Commentary confirms that the provision establishes a minimum humanitarian floor applicable across all forms of armed conflict regardless of the means employed.[5]

As operational systems become increasingly dependent on automated identification, autonomous classification, machine-assisted targeting, and algorithmic escalation processes, the practical implementation environment changes. The underlying obligation does not. The legal question is whether system architectures that systematically fail to process or preserve surrender-relevant indicators create heightened risks of non-compliance with obligations that have been operative for more than seven decades.

B. Additional Protocol I, Article 41: the duty of recognition stated in terms

Article 41(1) of Additional Protocol I provides: "A person who is recognized or who, in the circumstances, should be recognized to be hors de combat shall not be made the object of attack." Article 41(2) defines the protected condition: a person is hors de combat if he is in the power of an adverse Party, "clearly expresses an intention to surrender", or is incapacitated, provided he abstains from any hostile act and does not attempt to escape.[6]

The wording of paragraph 1 deserves more attention than it has received in the AI debate. The protection does not attach only to surrender that is in fact recognized; it attaches equally to surrender that, in the circumstances, *should be* recognized. The provision thereby imposes a normative standard of recognition on the attacking party — a duty to be capable of recognizing, calibrated to the circumstances. This is not an inference from the structure of the law; it is the text. Where the "circumstances" include engagement processes mediated by automated classification, the standard of what should be recognized is necessarily informed by what the deployed architecture is capable of recognizing. An architecture designed without any capacity to register the indicators of surrender does not lower the standard; it documents the distance between the standard and the system.

Article 41 operationalizes recognition through identifiable conditions and observable indicators. The ICRC Commentary explicitly notes that raised arms, abandonment of weapons, and verbal

[5]Geneva Conventions of 12 August 1949, common Article 3(1). See also ICRC, *Autonomous Weapon Systems and International Humanitarian Law: A Guide to the Issues* (2014); ICRC, *Autonomous Weapons Systems: Implications of Increasing Autonomy in the Critical Functions of Weapons* (Expert Meeting Report, 2023).

[6]Protocol Additional to the Geneva Conventions of 12 August 1949 (Protocol I), 8 June 1977, 1125 UNTS 3, Art. 41(1)–(2). Text as published by the UN Office of the High Commissioner for Human Rights. See Yves Sandoz, Christophe Swinarski & Bruno Zimmermann (eds.), *Commentary on the Additional Protocols of 8 June 1977* (ICRC/Martinus Nijhoff, 1987) paras 1593–1621.

surrender expressions may qualify as indicators regardless of the medium through which they are communicated.[7]

The question is therefore not whether AI systems become "subjects" of humanitarian law. The question is whether deployment architectures that systematically omit surrender-relevant recognition capabilities undermine the ability of human operators or operational chains to satisfy a recognition duty the treaty states in terms.

C. Additional Protocol I, Article 57: the duty of verification stated in terms

The recognition duty of Article 41 is reinforced by the precautionary regime of Article 57. Article 57(2)(a)(i) requires those who plan or decide upon an attack to "do everything feasible to verify that the objectives to be attacked are neither civilians nor civilian objects and are not subject to special protection." Verification is an information duty: it presupposes that status-relevant information — including hors de combat indicators — can reach the decision. Where the decision is reached through, or on the recommendation of, an automated system, the feasibility of verification is partly an architectural property: a system that cannot represent surrender-relevant states cannot supply them to the verifier.[8]

This reading is not confined to States party to the Protocol. The duty to do everything feasible to verify that targets are military objectives is customary (Rule 16 of the ICRC Study), as is the prohibition on attacking persons hors de combat (Rule 47) and the prohibition of denial of quarter (Rule 46; AP I Article 40), binding on all parties to a conflict regardless of treaty ratification status.[9]

An anticipated objection should be addressed here, because this is where it arises. It may be said that IHL imposes behavioural duties on human beings — to recognize, to verify, to refrain — and not technical capabilities on machines. The premise is correct; the conclusion does not follow, for two textual reasons. First, the precautionary regime itself extends beyond conduct in the moment of engagement: Article 57(2)(a)(ii) requires "all feasible precautions in the choice of means and methods of attack". The selection of the means is itself the object of a duty — and an engagement architecture is a means. Second, the recognition standard of Article 41(1) is indexed to "the circumstances", and the party that selects a machine-speed engagement process selects the circumstances in which recognition must occur. A party cannot rely on the impossibility of discharging a duty when that impossibility is the product of its own choice of means: the behavioural duty remains human, but its feasibility is determined upstream, at the level of design and procurement — which is precisely where Article 36 places the obligatory review. The argument therefore requires no transfer of legal personality to the machine; it requires only that duties about the choice of means be taken to mean what they say.

The convergence extends to the most significant non-party. The United States Department of Defense, in its July 2023 update to the *Law of War Manual*, added a dedicated section on "Feasible Precautions to Verify Whether Objects of Attack Are Military Objectives" and revised its guidance

[7] Sandoz et al., supra, para. 1602.
[8] AP I, Art. 57(2)(a)(i). See also Art. 57(2)(a)(ii) ("all feasible precautions in the choice of means and methods of attack") and ICRC, *Interpretive Guidance on the Notion of Direct Participation in Hostilities under IHL* (2009) 77–82.
[9] Henckaerts & Doswald-Beck, supra note 1, Rule 47, 166; see also Rule 46 (denial of quarter) and AP I, Art. 40.

to recognise a presumption of protected status absent information indicating a military objective. The verification duty, in other words, is stated in terms on both sides of the treaty boundary.[10]

D. Command Responsibility

The analysis of lawful operational safeguards intersects with the doctrine of command responsibility under Article 87 of Additional Protocol I and Article 28 of the Rome Statute. These provisions establish that commanders may incur individual responsibility for crimes committed by forces under their effective control where the commander knew or should have known of the risk and failed to take necessary and reasonable measures to prevent or repress such acts. Where a commander authorizes deployment of a system that systematically lacks surrender recognition capability — and where such capability was technically achievable and its absence reasonably foreseeable as a compliance risk — the failure to require such safeguards is a candidate "necessary and reasonable measure" not taken. As the temporal analysis in Section IV makes explicit, where the engagement window excludes meaningful human intervention, the architectural decision taken at procurement — rather than the operator's split-second choice — becomes the locus at which compliance is effectively determined.[11]

III. THE LEGAL REVIEW OBLIGATION: ARTICLE 36 AS PROCEDURAL ANCHOR

The duties examined in Section II are substantive. Additional Protocol I also contains the procedural provision that connects them to system architecture before deployment. Article 36 provides: "In the study, development, acquisition or adoption of a new weapon, means or method of warfare, a High Contracting Party is under an obligation to determine whether its employment would, in some or all circumstances, be prohibited by this Protocol or by any other rule of international law applicable to the High Contracting Party."[12]

Three features of this obligation carry the argument. First, its temporal scope: the review attaches to "study, development, acquisition or adoption" — the ICRC Guide confirms that this covers all stages of the procurement process, from conception onward. Article 36 therefore operates at exactly the point where architectural choices, including the inclusion or exclusion of recognition capabilities, are made and are still reversible. Second, its material scope: the review must assess compatibility not only with the Protocol's specific weapons prohibitions but with "any other rule of international law applicable" — which includes the recognition duty of Article 41(1) and the verification duty of Article 57(2)(a)(i) examined above. Third, its mandatory character: Article 36 is not a recommendation; it is "an obligation".[13]

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[10] US Department of Defense, *Law of War Manual*, July 2023 update, § 5.5.3 ("Feasible Precautions to Verify Whether Objects of Attack Are Military Objectives") and revised § 5.4.3, recognising a presumption of protected status absent information indicating a military objective.

[11] On command responsibility and autonomous systems, see Schmitt & Thurnher, supra note 2, 260–265; Marco Sassoli, 'Autonomous Weapons and International Humanitarian Law' (2014) 90 International Law Studies 308; Human Rights Watch & IHRC, *Mind the Gap* (2015).

[12] AP I, Art. 36. Text as published by OHCHR.

[13] ICRC, *A Guide to the Legal Review of New Weapons, Means and Methods of Warfare: Measures to Implement Article 36 of Additional Protocol I of 1977* (Kathleen Lawand ed., 2006), noting that the temporal application of Article 36 "covers all stages of the weapons procurement process", from conception and study through development to acquisition and adoption.

The consequence can be stated plainly. Whether an AI-enabled operational system is capable of identifying, preserving, or escalating surrender-relevant indicators within its engagement window is not a question this paper invents and offers for consideration. It is a question that the legal review which States are already obliged to conduct is already required to ask, because the employment of a system incapable of supporting recognition and verification "in some or all circumstances" is precisely what the review exists to determine. The contribution of this paper is to make the question explicit and to supply criteria — set out in Sections V and VII — by which it can be answered in a documented, auditable form.

Nor can the novelty of the technology displace the inquiry. The International Court of Justice, in its 1996 Advisory Opinion, held that the established principles and rules of humanitarian law apply "to all forms of warfare and to all kinds of weapons, those of the past, those of the present and those of the future." The temporal openness of the law is judicially settled; Article 36 is its procedural expression.[14]

IV. THE TEMPORAL FOUNDATION: HUMAN DECISION-MAKING AND MACHINE-SPEED OPERATIONS

The analysis developed in this paper rests upon a premise that is often acknowledged but insufficiently examined: the problem of time. International Humanitarian Law presumes the existence of decision-makers capable of assessing circumstances, interpreting observable conduct, and applying legal obligations before force is employed. Historically, these functions have been performed by human actors operating within biological and cognitive constraints.

A. Human Decision-Making Within Biological Limits

Human cognition operates within measurable temporal limits. Experimental research places basic perceptual processing on the order of one hundred milliseconds, with motor response adding a further fifty to one hundred milliseconds; simple reaction times generally fall between roughly 150 and 300 milliseconds. These figures, however, describe simple reaction — a single pre-cued response to an expected stimulus. The decision relevant to the lawful use of force is categorically different. It requires perception, recognition, discrimination between combatant and hors de combat status, evaluation of alternatives, and a judgment to act or to refrain. Such choice-and-judgment responses extend substantially beyond the simple-reaction range and increase with task complexity.[15]

This distinction is particularly significant in military and security environments, and the operational context tends to degrade rather than improve performance. Acute stress, cognitive load, fatigue, and environmental ambiguity have measurable effects on both reaction time and decision quality. Experimental work on lethal-force decision-making confirms that engagement and restraint decisions cannot be reduced to reflexive reactions: they are measurable in the hundreds-of-milliseconds-to-seconds range and are sensitive to stressor conditions. The human decision to use, or to withhold, lethal force is therefore not a sub-second act; it is a deliberative process whose

[14] *Legality of the Threat or Use of Nuclear Weapons* (Advisory Opinion) [1996] ICJ Rep 226, para. 86.
[15] On the distinction between simple, recognition, and choice reaction times, see R. J. Kosinski, 'A Literature Review on Reaction Time' (Clemson University, updated 2013); W. E. Hick, 'On the Rate of Gain of Information' (1952) 4 Quarterly Journal of Experimental Psychology 11; R. Ratcliff & G. McKoon, 'The Diffusion Decision Model' (2008) 20 Neural Computation 873.

realistic floor lies well above the simple-reaction range and whose practical duration, under operational stress, may extend to seconds.[16]

B. Machine-Speed Operations

Autonomous and AI-assisted systems operate within a fundamentally different temporal domain. Computational processes associated with detection, classification, prioritization, targeting support, and engagement recommendation may occur within milliseconds or seconds, depending upon system architecture and operational design. Defensive systems such as Phalanx and SeaRAM complete their detection, evaluation, and response sequence within a matter of seconds, and once activated leave little practical room for human intervention.[17]

More generally, a system that classifies an object as a potential threat may initiate an engagement within milliseconds — long before a human operator has the opportunity to assess the situation or intervene effectively. As autonomy increases, these cycles contract further; commentators observe that calculations performed at machine speed may eclipse any horizon for timely, meaningful human intervention.[18]

C. Temporal Asymmetry and the Compression of Human Control

The significance of this disparity does not depend upon any single numerical threshold. What remains constant is the existence of a structural temporal asymmetry between biological cognition and computational processing: the machine's decision-to-action cycle may operate within a window — milliseconds to a few seconds — that falls below the threshold required for meaningful human assessment of the engagement. This phenomenon has increasingly been described as temporal compression. As operational systems become faster, the time available for human intervention contracts; decision windows that once permitted independent human assessment may shrink to the point where human participation becomes largely procedural. Under such conditions, the mere presence of a human operator within the operational chain does not establish meaningful human control: where machine-generated outputs are produced faster, and in greater volume, than they can realistically be assessed, operators become dependent upon automated recommendations, with the documented risks of automation bias and "rubber-stamping". A human may remain technically "in the loop" while the loop itself runs faster than human cognition.[19]

The implications for International Humanitarian Law follow directly. If the practical opportunity for independent human evaluation diminishes as system speed increases, compliance with legal obligations cannot rely exclusively upon external human intervention; compliance-relevant functions must be preserved within the system architecture itself. The argument advanced in this paper proceeds from this observation: surrender recognition capability is examined not as an ethical preference, but as a lawful operational safeguard capable of preserving compliance-relevant information within decision environments characterized by temporal compression. The relevant

[16] B. Heusler & C. Sutter, 'Shoot or Don't Shoot?' (2022) 13 Frontiers in Psychology 798766; 'Oscillatory Neural Correlates of Police Firearms Decision-Making in Virtual Reality' (2024) 11 eNeuro; L. Bartel et al., 'Time to Stop: Firearm Simulation Dynamics' (Force Science / VirTra, 2025).

[17] 'Against the Growing Anti-Ship Missile Threat', Center for International Maritime Security (2016), citing NavWeps data on Phalanx CIWS automatic-mode engagement sequences.

[18] Scandinavian Journal of Military Studies (2026), summarizing M. Mazarr et al. (2020).

[19] ICRC Humanitarian Law & Policy Blog, 'The (im)possibility of meaningful human control for lethal autonomous weapon systems' (29 August 2018).

question is not whether autonomous systems should be faster or slower than human operators, but whether operational architectures preserve sufficient opportunity for compliance with existing legal obligations when machine decision cycles outpace meaningful human assessment.[20]

The structure of this argument — a duty whose exercise presupposes a humanly bridgeable time window, a class of systems that closes the window, and the consequent migration of the duty into design — is not unique to humanitarian law.[21]

V. SURRENDER RECOGNITION AS A CAPABILITY CLASS

A. Definition and Technical Feasibility Criteria

For purposes of this paper, surrender recognition capability refers to any technical mechanism in an operational AI system designed to: identify surrender expressions; recognize de-escalation indicators; preserve uncertainty during ambiguous encounters; inhibit automatic escalation under uncertainty; escalate ambiguous cases to human review; and maintain operational awareness of hors de combat conditions.

The concept does not require perfect recognition. Human operators themselves routinely operate under uncertainty; IHL does not demand infallibility. The relevant inquiry concerns whether foreseeable and technically achievable mitigation mechanisms are systematically excluded. The term “technically achievable” is not an abstract standard. For purposes of procurement and audit analysis, feasibility is assessed against verifiable criteria including: Technology Readiness Level (TRL) of relevant sensor and classification components, where TRL 6 or above may indicate sufficient maturity for procurement consideration; acceptable inference latency thresholds for the operational tempo of the system in question; availability of annotated training datasets for surrender-relevant signal categories in representative environments; and capacity for independent validation of recognition performance against defined recall and precision benchmarks. Established frameworks such as the NIST AI Risk Management Framework and the IEEE 7000 series provide further reference criteria for structured feasibility assessments. These criteria are offered as anchors for a documented assessment, not as thresholds this paper purports to fix: whether the line sits at TRL 6 or 7, or what recall is adequate for a given tempo, are determinations for the reviewing authority — the legal point is that the determination must be made, documented, and reviewable, rather than left implicit in a procurement silence.[22]

The latency criterion is the direct operational expression of the temporal analysis in Section IV: for a safeguard to be compliance-relevant, its inference latency must fall within the decision window of

[20] K. L. Mosier & L. J. Skitka, ‘Human Decision Makers and Automated Decision Aids’ (1996), and the tempo-compression discussion in the autonomous-weapons scholarship cited supra notes 18–19.

[21] The temporal structure analysed in this Section is not unique to IHL. For the parallel argument in civil machine-safety law — where the stop-function, emergency-stop, and guarding duties of Directive 2006/42/EC, Regulation (EU) 2023/1230, the Swiss LSPro and OSHA 29 CFR § 1910.212 rest on the same implicit assumption of a humanly bridgeable interval — see G. Nardacci, ‘Systemic Arrestability: Operator Protection in Machine Safety Law When Machine Speed Exceeds Human Reaction’ (Human Flag Association working paper, 2026), SSRN Abstract ID 6923461, https://papers.ssrn.com/abstract=6923461. The two analyses are independent and rest on separate legal anchors; their convergence indicates that the effectiveness problem identified here is structural rather than regime-specific.

[22] National Institute of Standards and Technology, *AI Risk Management Framework* (AI RMF 1.0) (NIST AI 100-1, 2023); IEEE 7000-2021; NATO STO, *Technology Readiness Levels in NATO Science and Technology* (2019); UK MoD, *Ambitious, Safe, Responsible* (2022).

the system in which it operates. A surrender-recognition mechanism that operates at machine speed addresses precisely the gap that human oversight, under temporal compression, cannot fill. This framing draws on the principle of precaution in attack reflected in Article 57: where surrender recognition constitutes a technically achievable precautionary mechanism against the criteria above, its systematic exclusion requires justification within the feasible-precaution analysis — and within the Article 36 review of Section III.

B. Technical Forms

Potential implementation forms include: multimodal surrender signal recognition; gesture recognition; linguistic surrender detection; uncertainty escalation thresholds; human override escalation pathways; refusal-to-engage constraints under unresolved surrender ambiguity; and probabilistic de-escalation weighting. This paper intentionally avoids endorsing any single implementation architecture. The legal relevance arises not from a particular model design, but from the operational function served.

C. Operational Relevance

In operational environments, escalation errors occur not only through aggression, but through the absence of de-escalation mechanisms. A system optimized exclusively for hostile detection without surrender-sensitive mitigation pathways increases: false hostile continuation; escalation persistence; failure to preserve surrender ambiguity; and downstream attribution risk. Under such circumstances, the absence of safeguards does not increase operational safety or national security; in some environments, it foreseeably reduces both.[23]

D. Illustrative Operational Scenarios

The following scenarios are illustrative only, intended to concretize the compliance risk analysis rather than to assert factual conclusions about any specific system or deployment.

Scenario 1 — ISR drone in dense urban environment. An ISR drone operating in a densely populated urban environment is equipped with an automated threat classification system, optimized to identify hostile actors and flag targets for human-authorized engagement. It includes no mechanism for detecting or flagging surrender-relevant behaviors — raised hands, discarded weapons, stationary posture following prior movement — as a distinct classification state. In an engagement sequence, a combatant who has ceased hostile activity and raised hands is classified as "still present in threat zone". The system provides no signal to the human operator that a potentially hors de combat condition may exist. The operator, relying on the automated classification, proceeds with engagement authorization. The architecture has foreclosed a compliance-relevant decision pathway before the human operator was engaged — and, in the language of Article 41(1), has shaped the "circumstances" in which surrender should have been recognized. Whether a legally relevant failure occurred depends on facts not assumed here; what the scenario establishes is the locus of the problem: design, not operator judgment.

Scenario 2 — Decision support system for indirect fire. A decision support system for indirect fire operations processes sensor fusion data to provide targeting recommendations to a fire mission

[23] This observation is consistent with scholarship on escalation dynamics in autonomous systems. See Paul Scharre, *Army of None* (W.W. Norton, 2018) ch. 14; RAND Corporation, *Human-Machine Teaming for Future Ground Forces* (2020).

authority, ranking potential targets by threat probability. It includes no uncertainty escalation logic for scenarios in which available data is insufficient to distinguish active hostile engagement from cessation of hostilities or withdrawal. A fire mission authority, operating under time pressure that reflects the temporal compression identified in Section IV, authorizes engagement without independent assessment of hors de combat status. Post-engagement review reveals that sensor data available at the time of the recommendation contained indicators consistent with cessation of hostile activity that the system's architecture was not designed to process or flag. The absence of uncertainty escalation logic has reduced the practical operability of the verification duty under Article 57(2)(a)(i); the procurement decision not to include such logic is exactly the kind of determination the Article 36 review exists to examine.

VI. LAWFUL OPERATIONAL SAFEGUARDS

A. Distinguishing Policy Safeguards from Compliance-Relevant Safeguards

Public and regulatory discussions frequently treat all AI safeguards as belonging to a single undifferentiated category. This obscures an important distinction. Some safeguards are discretionary policy constraints reflecting corporate preference, reputational concern, or product strategy. Others perform functions directly relevant to legal compliance. The latter category can now be defined: *a lawful operational safeguard is a technically implementable system function whose absence foreseeably degrades compliance with an existing legal obligation within the decision window in which operational outcomes are determined.* Each element of the definition does work: "technically implementable" excludes the aspirational; "foreseeably degrades" anchors the category in established risk analysis rather than hindsight; "existing legal obligation" excludes preference and policy; and "within the decision window" incorporates the temporal criterion of Section IV, distinguishing safeguards that operate where the decision is actually made from oversight that arrives after it. Such safeguards are not mandated in explicit technical form by existing treaties; they are the foreseeable mitigation mechanisms through which preexisting legal obligations — recognition, verification, precaution — remain capable of being discharged in machine-mediated decision environments.

B. Compliance Architecture

Modern operational systems increasingly depend upon algorithmic classification, probabilistic targeting, autonomous assistance, machine-mediated escalation, and automated operational filtering. As a result, legal compliance increasingly interacts with system architecture. This interaction is reinforced by temporal asymmetry: as operational decision cycles accelerate, legal compliance cannot rely exclusively on post hoc human intervention; certain compliance-relevant functions must be preserved within the architecture itself, because the available decision window is insufficient for meaningful human review. Under this framework, such safeguards function not as "ethical add-ons" but as compliance-preserving mechanisms, operational risk mitigations, uncertainty preservation structures, and escalation-limiting controls. Their legal significance arises not from moral aspiration, but from operational consequence.

C. Removal as Legal Risk

Where technically achievable safeguards relevant to surrender recognition are intentionally excluded from operational AI systems, foreseeable risks arise: increased escalation under uncertainty; increased probability of failure to recognize hors de combat conditions; reduced human review opportunities; and weakened operational restraint pathways. Removal of such safeguards is therefore not a neutral operational choice; it is a design determination that engages the duties examined in Sections II and III and warrants documented justification. This does not imply automatic illegality. It means that safeguard absence is relevant to legal and operational evaluation — including command responsibility analysis, as discussed in Section II.D, and Article 36 review, as discussed in Section III.[24]

For audit and procurement purposes, systematic exclusion may be evidenced by one or more of the following indicators: procurement specifications that explicitly exclude surrender-relevant signal classes from system requirements; absence of hors de combat or de-escalation scenarios in pre-deployment validation protocols; technical documentation or supplier statements indicating that surrender recognition is not a supported or intended capability; or architectural design choices that remove or preclude human escalation pathways in ambiguity-resolving contexts.

VII. TECHNICAL LIMITATIONS AND BOUNDARY CONDITIONS

Any credible analysis of surrender recognition as a compliance-relevant capability must acknowledge significant technical limitations. These limitations do not defeat the argument; they define its proper scope.

A. False Positives, Adversarial Signaling, and Robustness

Recognition systems may generate false positives, incorrectly classifying non-surrender gestures or communications as surrender indicators. In adversarial environments, actors may deliberately exploit surrender recognition mechanisms to create operational pauses, gain tactical advantage, or manipulate escalation dynamics. Research on adversarial examples in deep learning demonstrates that small, deliberately crafted perturbations can cause classification systems to produce incorrect outputs with high confidence; gesture spoofing, acoustic mimicry, and signal injection are real vulnerabilities and must be addressed through structured red-teaming, adversarial scenario testing, and robustness validation as part of any pre-deployment audit process.[25]

However, the existence of adversarial risk does not resolve the underlying legal question. IHL itself operates under conditions of deception — it prohibits perfidy (Article 37) precisely because false surrender is a known tactic — and it does not suspend obligations because of the risk of false signals. The relevant question is whether the system architecture provides any meaningful pathway for surrender-relevant information to influence operational outcomes, and whether the system has been tested against adversarial conditions as part of a structured assurance process.

Recognition systems may also generate false negatives — failing to identify genuine surrender indicators due to sensor limitations, model underfitting, or threshold calibration optimized for precision over recall. In IHL terms, a false negative directly implicates the verification duty of Article 57(2)(a)(i), as it increases the risk of attacking persons hors de combat. Where operational systems are calibrated to minimize false positives at the expense of recall, the resulting degradation

[24] On foreseeable risk as a basis for operational legal analysis, see ICRC *Interpretive Guidance*, supra note 8, 77–82.

[25] Christian Szegedy et al., ‘Intriguing Properties of Neural Networks’ (2014) ICLR; Ian J. Goodfellow, Jonathon Shlens & Christian Szegedy, ‘Explaining and Harnessing Adversarial Examples’ (2015) ICLR; Nicholas Carlini & David Wagner, ‘Towards Evaluating the Robustness of Neural Networks’ (2017) IEEE S&P 39.

in surrender-signal detection requires specific justification under feasible precaution analysis. This does not imply that perfect recall is required; it means that the calibration rationale should be documented and reviewed as part of compliance-relevant assurance processes.

B. Battlefield Ambiguity and Imperfect Recognition Environments

Operational environments present conditions that may substantially degrade recognition performance: low visibility, sensor degradation, partial data, communication interference, and high-tempo decision chains. These conditions may constrain the practical implementation of surrender recognition mechanisms in certain operational contexts. They do not eliminate the analytical relevance of the capability class. The principle of feasible precaution under Article 57 is explicitly conditioned on what is achievable under the circumstances; where recognition capability is not feasible given the TRL and operational parameters of the system, the analysis changes accordingly. Even where uncertainty cannot be eliminated, the temporal asymmetry of Section IV remains relevant: the challenge is not only recognition accuracy but the preservation of sufficient decision space for lawful assessment. The paper does not claim that surrender recognition capability is universally achievable. It claims that its systematic exclusion from systems where it is technically feasible — assessed against the criteria in Section V.A — engages the duties of recognition, verification, and review.

C. Scope Limitation: Operational Systems Only

The analysis in this paper is confined to AI systems with operational, targeting, escalation, or battlefield-relevant functions. It does not apply to consumer AI, general-purpose language models, logistics systems, or administrative applications. This scope limitation is analytically significant: the compliance-relevance of surrender recognition capability arises specifically from the operational consequence of its absence in systems capable of influencing kinetic or escalatory outcomes.

VIII. IMPLICATIONS FOR FUTURE GOVERNANCE

A. Autonomous Systems and International Governance

As AI systems become increasingly integrated into operational, military-adjacent, intelligence, and escalation-sensitive environments, the distinction between discretionary alignment restrictions and lawful operational safeguards will become increasingly relevant in autonomous systems governance, dual-use AI regulation, military AI procurement, operational accountability analysis, and IHL adaptation debates — including the GGE LAWS process.

B. European Regulatory Context and Dual-Use Governance

The EU Artificial Intelligence Act expressly excludes AI systems developed or used exclusively for military purposes from its scope under Article 2(3); the framework proposed here makes no claim of direct applicability of the Act to military AI systems. The European regulatory landscape remains relevant through two channels. First, dual-use AI systems may fall within the Act’s high-risk classification under Annex III, in which case its requirements for risk management, technical documentation, and human oversight intersect with the safeguard analysis proposed here. Second, European and NATO member state procurement frameworks are subject to national standards, dual-use export controls, and interoperability requirements that create de facto compliance obligations; the concept of foreseeable misuse as a risk analysis criterion engages compliance-relevant questions under both IHL and European risk governance frameworks even where the AI Act does not directly apply.[26]

C. Procurement Compliance Framework

The procurement framework proposed in this section maps onto existing legal and institutional review processes — first and foremost the Article 36 review examined in Section III, which provides its natural procedural home. NATO AI assurance initiatives and national frameworks such as the UK MoD's AI approach and US DoD Directive 3000.09 increasingly emphasize structured testing, documentation, and human oversight criteria that align with the checklist elements below. The following framework is offered as an operational checklist for acquisition authorities, certification bodies, and independent auditors evaluating operational AI systems with escalation-relevant functions.

1. Technical Feasibility Assessment. Has surrender recognition capability been assessed as technically feasible for this system in its intended operational environment? Has the assessment been documented with reference to TRL criteria, sensor specifications, and latency constraints? Has it been conducted or reviewed independently of the system developer?

2. Exclusion Justification. Where surrender recognition capability has been excluded, has the exclusion been documented with specific operational justification? Has the justification addressed whether the exclusion is permanent or context-specific? Has the foreseeable IHL compliance risk created by the exclusion been documented and reviewed?

3. Human Escalation Pathways. Does the system architecture include a "pause for verification" pathway triggered by surrender-ambiguous indicators? Is human escalation available before irreversible action in unresolved de-escalation scenarios, and is it feasible within the system's decision window (Section IV)? Are escalation pathways documented in operational procedures and tested under realistic conditions?

4. Pre-Deployment Audit. Has an independent pre-deployment audit been conducted using surrender and de-escalation scenarios? Have adversarial signaling and spoofing scenarios been included in the audit test suite? Has audit documentation been retained for post-deployment operational review?

5. Command Responsibility Documentation. Has authorizing command reviewed and acknowledged the IHL compliance risk profile of the system? Is there a documented record of the feasible precaution analysis — and of the Article 36 review — conducted prior to deployment authorization?

IX. LIMITATIONS OF THE ARGUMENT

This paper does not claim: that current international law prohibits AI systems; that all autonomous systems are unlawful; that surrender recognition is technologically solved; that any single safeguard architecture is legally mandatory; or that operational uncertainty can be eliminated.

[26] Regulation (EU) 2024/1689 (EU Artificial Intelligence Act), Art. 2(3), Annex III.

The argument is determinate but bounded. It establishes that existing IHL obligations — recognition under Article 41(1), verification under Article 57(2)(a)(i), legal review under Article 36 — remain operative in AI-mediated operational environments and are stated in the treaty text in terms; that the temporal compression of decision cycles can render human oversight nominal rather than substantive; that certain safeguard categories serve compliance-relevant operational functions; and that the systematic exclusion of surrender-sensitive safeguards from systems where they are technically achievable — assessed against verifiable TRL and operational criteria — engages those duties and requires documented justification. The paper neither advocates categorical AI restrictions nor accepts the proposition that fewer safeguards are equivalent to greater operational capability or legal safety. The translation of IHL principles into technical design specifications raises complex questions that remain unresolved in both legal scholarship and technical practice; the argument presented here is offered as a contribution to that discussion, not as a resolution of it.

KEY TAKEAWAYS FOR PROCUREMENT AND COMMAND STAFF

1. Function — Lawful operational safeguards preserve compliance-relevant information. They are not optional ethical add-ons. Their presence or absence has operational and legal consequences.

2. Law — The duties are textual: recognition (“should be recognized”, Art. 41(1) AP I), verification (“everything feasible to verify”, Art. 57(2)(a)(i)), and legal review at procurement (Art. 36).

3. Time — Where machine decision cycles outpace human reaction, oversight becomes nominal. Confirm that any human-review pathway is feasible within the system’s actual decision window.

4. Feasibility — Assess surrender recognition capability against verifiable criteria: TRL ≥6, acceptable latency thresholds, annotated training datasets, independent validation. Document the assessment.

5. Documentation — Record exclusions, operational justifications, and feasible precaution analysis. Retain documentation for audit, command review, and Article 36 files.

6. Accountability — Integrate safeguard analysis with command responsibility doctrine. Awareness of foreseeable compliance risk is a prerequisite of the ‘necessary and reasonable measures’ standard.

X. CONCLUSION

The framing that fewer safeguards necessarily increase operational effectiveness or national security is incomplete. Certain safeguards function as operational mechanisms through which existing humanitarian obligations remain capable of being discharged — and their absence is a fact the law already has the instruments to examine.

The compliance challenges associated with autonomous systems are not solely questions of accuracy or capability. They are questions of time. Human judgment operates within biological limits, while computational systems increasingly operate at machine speed. Where this temporal asymmetry becomes sufficiently pronounced, meaningful human control becomes nominal rather than substantive, and compliance-relevant functions must be preserved within the architecture, because the architecture is the only actor present within the window in which the decision is actually made.

Surrender recognition capability in operational AI systems is the paradigm case of such a lawful operational safeguard. The duty to recognize surrender that “should be recognized” is in the text of Article 41. The duty to do “everything feasible to verify” is in the text of Article 57. The duty to examine new means and methods of warfare against those obligations, at the stage when architecture is decided, is in the text of Article 36. The International Court of Justice has confirmed that these principles govern the weapons of the future no less than those of the past. What this paper adds is the connection: in machine-speed decision environments, those textual duties are discharged or defeated at the level of system design — and a procurement process that never asks whether the system can recognize surrender has not answered the question; it has only avoided asking it.

If the analysis presented here is correct, the relevant question in future governance, procurement, and accountability discussions is not whether AI systems should have fewer or more safeguards as a general matter. It is which safeguards perform compliance-relevant functions under existing law, what verifiable criteria determine their technical feasibility, and how the sequence technical feasibility → documented justification → command awareness → operational accountability structures legal review under obligations that already bind.

METHODOLOGICAL NOTE

This paper proposes an analytical framework grounded in treaty text; it does not purport to resolve disputed questions of treaty interpretation, nor to establish binding standards of conduct beyond those the cited provisions themselves impose. The qualifications that earlier editions distributed throughout the text are consolidated here: the paper advances no claim that any specific technical implementation is legally mandated; the illustrative scenarios in Section V.D are analytical constructs and do not describe or evaluate any existing system or deployment; the temporal figures in Section IV are drawn from published literature and are presented as orders of magnitude establishing a structural asymmetry, not as operational specifications; and references to procurement frameworks and audit criteria are analytical reference points, not legal advice. The author welcomes engagement, critique, and further development of the framework by scholars, practitioners, and policymakers working in international humanitarian law, autonomous systems governance, and defence procurement.[27]

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[27] All treaty citations in this edition have been verified against the official texts (OHCHR instrument database; ICRC IHL databases) as of June 2026. They remain subject to confirmation by qualified counsel prior to formal submission.

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